

Event-Driven Notification Dispatcher Architecture Diagram

Flow

Client (Postman / Browser)
↓
POST /api/v1/events
↓
Express.js API (eventController.js)
↓ Validate Request
↓
Event Service (eventService.js)
↓
Save Event → SQLite **events** table
↓
Create Notification (status = pending) → SQLite **notifications** table
↓
Push Notification Task → In-Memory Queue (JavaScript Array)
↓
Return **HTTP 202 Accepted** immediately to the client
↓
Background Queue Worker (queueWorker.js)
↓
Random Delay (500–1000 ms)
↓
Simulate Notification Sending (10% Failure Rate)
↓
Update Notification Status → completed / failed
↓
Increment retry_count on failure

Immediate API Response

```
{  
  "message": "Event accepted for processing",  
  "tracking_id": 1,  
  "notification_id": 1,  
  "status": "pending"  
}
```

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